

ASSIGNMENT-1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples

Answer:-

Frontend Development:

Frontend is the part of a website or app that users can see and use. It includes buttons, images, menus, and pages.

Example: On Amazon, the homepage, product images, and shopping cart are frontend.

Backend Development:

Backend is the part that works behind the scenes. It stores data, checks login details, and processes orders.

Example: When you log in to Amazon, the backend checks your username and password.

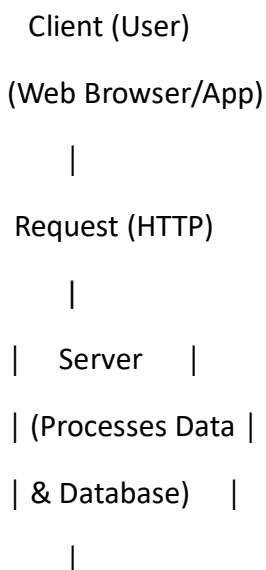
Full-Stack Development:

A full-stack developer works on both the frontend and backend. They can build a complete website or application.

Example: A full-stack developer can create a food delivery app with both the user interface and the server.

2. Create a simple diagram showing how the client-server model works in web architecture.

Answer:-



Response (HTML/

CSS/JS/Data)

|

Client (User)

Displays the Result

3. Describe how a browser requests and displays a web page from a web server.

Answer:-

How a Browser Requests and Displays a Web Page

1. The user enters a website address (URL) in the browser.
2. The browser sends a request to the **web server**.
3. The server processes the request and finds the required web page.
4. The server sends the page files (HTML, CSS, JavaScript) back to the browser.
5. The browser reads these files and displays the web page on the screen.

Example: When you type www.google.com, the browser sends a request to Google's server. The server returns the webpage, and the browser displays it for you.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Answer:-

Tool	Purpose
Code Editor (VS Code)	Used to write and edit HTML, CSS, and JavaScript code.
Web Browser (Chrome/Firefox)	Used to run and test web pages.
Git	Helps track changes in code and manage projects.
GitHub	Stores code online and allows collaboration with others.
Node.js	Runs JavaScript outside the browser and installs packages.

Tool	Purpose
npm (Node Package Manager)	Installs and manages libraries and tools for web development.

5. Explain what a web server is and give examples of commonly used servers.

Answer:-

What is a Web Server?

A **web server** is a software or computer that stores website files and sends them to users' browsers when they request a webpage.

Examples of Common Web Servers:

- **Apache HTTP Server** – One of the most widely used web servers.
- **Nginx** – Fast and efficient, commonly used for high-traffic websites.
- **Microsoft IIS (Internet Information Services)** – Used mainly on Windows servers.
- **LiteSpeed** – Known for its high performance and speed.

Example:

When you open **www.amazon.com**, your browser sends a request to Amazon's web server. The server sends the webpage back, and the browser displays it.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Answer:-

Roles of Frontend Developer, Backend Developer, and Database Administrator

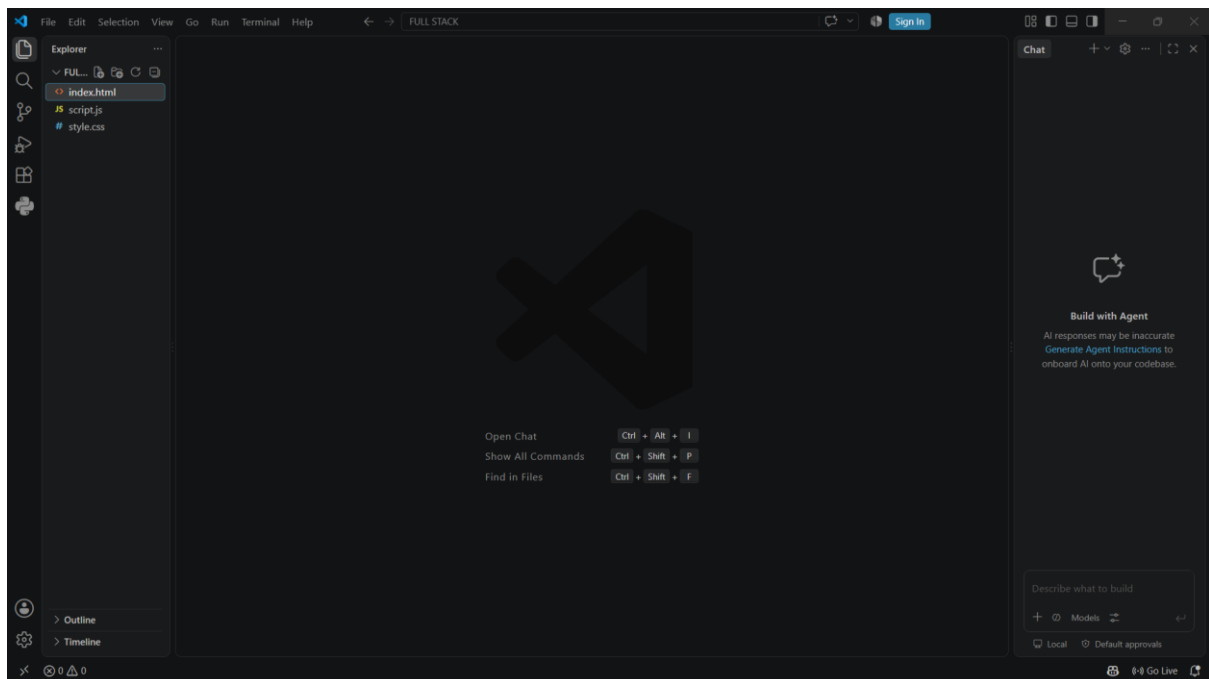
- **Frontend Developer:**
Creates the part of the website or app that users see and interact with, such as buttons, menus, and pages.
- **Backend Developer:**
Builds the server-side logic, handles user requests, manages data, and connects the application to the database.
- **Database Administrator (DBA):**
Manages the database, stores data securely, performs backups, and ensures data is organized and easily accessible.

Example

In an online shopping website:

- **Frontend Developer** designs the product pages and shopping cart.
- **Backend Developer** processes user logins and orders.
- **Database Administrator** stores customer, product, and order information securely.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.



8. Explain the difference between static and dynamic websites. Provide an example of each.

Answer:-

Difference Between Static and Dynamic Websites

Static Website

Shows the same content to every user.

Uses only HTML and CSS.

Simple and faster.

Dynamic Website

Content can change based on the user or data.

Uses HTML, CSS, JavaScript, and backend languages with a database.

More interactive and feature-rich.

Examples

- **Static Website:** A personal portfolio or company information website.
- **Dynamic Website:** Amazon, Facebook, or Instagram, where content changes for each user.

9. Research and list five web browsers. Explain how rendering engines differ between them.

Answer:-

Five Web Browsers and Their Rendering Engines

Web Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Microsoft Edge	Blink
Apple Safari	WebKit
Opera	Blink

How Rendering Engines Differ

A **rendering engine** is the part of a browser that reads HTML, CSS, and JavaScript and displays the webpage.

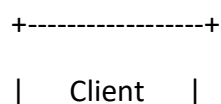
- **Blink:** Fast and used by Chrome, Edge, and Opera.
- **Gecko:** Used only by Firefox and has its own way of displaying web pages.
- **WebKit:** Used by Safari and is optimized for Apple devices.

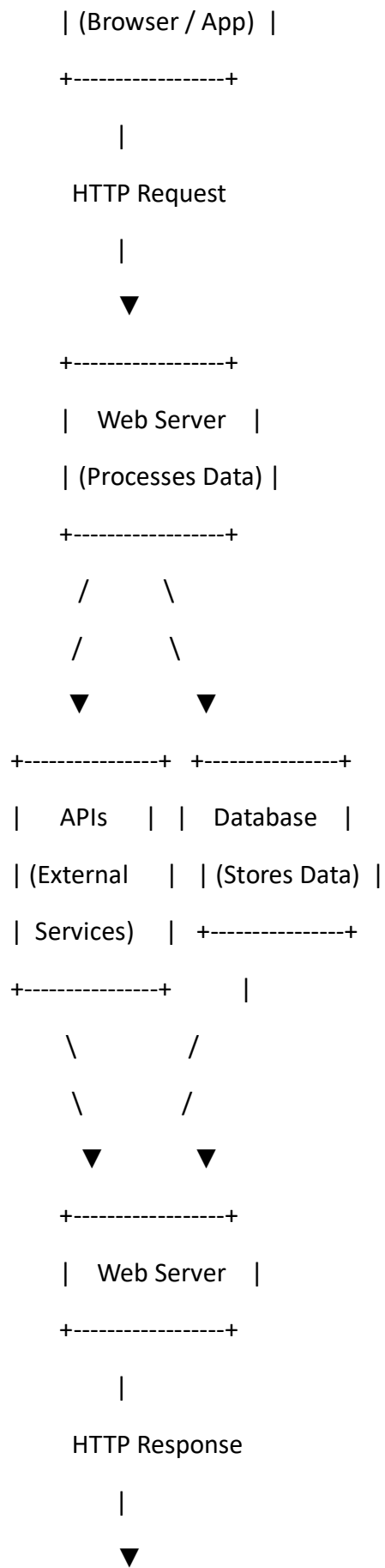
Example:

The same website may look or behave slightly differently in Chrome, Firefox, and Safari because each browser uses a different rendering engine.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Answer:-





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| Client |

| Displays Result |

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